

Some interesting problems in analysis, probability, and statistics

Collected mostly from my undergraduate degree in applied math at McGill (& EPFL)

Credit for many problems goes to professors Toth, Chen, Dalang, Lin, Khalili, and Nešlehová.

Notes: In general, the problems go deeper into the field through each section. I have chosen these problems not because they are necessarily particularly tricky, but because they include some interesting ideas that I was happy to see touched on when I encountered them. Please contact me if you want more hints or a solution, though plenty are available online. The 'probability and stochastic processes' section is my favourite.

Problems in calculus & analysis

1. Let g be a continuous real-valued function on the unit circle $\{x \in \mathbb{R}^2 : |x| = 1\}$ such that $g(0, 1) = g(1, 0) = 0$ and $g(-x) = -g(x)$.

Define $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ by $f(x) = 0$ if $x = 0$ and $f(x) = g\left(\frac{x}{|x|}\right)|x|$ otherwise.

- (a) Let $x \in \mathbb{R}^2$. Define $h : \mathbb{R} \rightarrow \mathbb{R}$ by $h(t) = f(tx)$. Show that h is differentiable.
 - (b) Show that f is not differentiable at $(0, 0)$ unless $g = 0$ (g is the zero function).
2. Let $f : [0, 1] \times [0, 1] \rightarrow \mathbb{R}$ be given by:

$$f(x, y) = \begin{cases} 2^{2n} & \text{for } (x, y) \in [2^{-n}, 2^{-n+1}) \times [2^{-n}, 2^{-n+1}), n \in \mathbb{N} \\ -2^{2n+1} & \text{for } (x, y) \in [2^{-n-1}, 2^{-n}) \times [2^{-n}, 2^{-n+1}), n \in \mathbb{N} \\ 0 & \text{otherwise} \end{cases}$$

Prove or disprove the following statement:

$$\int_0^1 \int_0^1 f(x, y) dx dy = \int_0^1 \int_0^1 f(x, y) dy dx$$

Hint: first integrate in one variable, then the other.

3. Define

$$I(\alpha) = \int_0^\infty \ln(1 + \alpha e^{-x} + e^{-2x}) dx.$$

By finding $\frac{dI}{d\alpha}$, show that $\sum_{n=1}^\infty \frac{1}{n^2} = \frac{\pi^2}{6}$. Justify your answer carefully.

Hint: to simplify the integral, consider only the range $-2 \leq \alpha \leq 2$. Use the fact that $I(2) = 4I(0)$. To conclude, find an expression for $I(-2)$ and apply Taylor's theorem. This formula may be useful:

$$\arctan(x) + \arctan(y) = \arctan\left(\frac{x+y}{1-xy}\right) \quad \text{when } xy \neq 1$$

4. Let $\varepsilon > 0$ be fixed. Show that $\text{Leb}(\mathcal{X}) = 0$, where $\mathcal{X}$ is the set of all real numbers $x \in [0, 1]$ such that there exist infinitely many pairs $(p, q) \in \mathbb{N}^2$ with

$$\left|x - \frac{p}{q}\right| < \frac{1}{q^{2+\varepsilon}}$$

5. Construct a sequence of functions $\{f_n\}_{n=1}^\infty, f_n : \mathbb{R} \rightarrow [0, \infty)$, with $f_n \geq 0$ and measurable $\forall n \geq 1$ such that

$$\int_{\mathbb{R}} \liminf_{n \rightarrow \infty} f_n d(\text{Leb}) < \liminf_{n \rightarrow \infty} \int_{\mathbb{R}} f_n d(\text{Leb}).$$

Problems in probability theory & stochastic processes

6. Let $X \sim \text{Cauchy}(0, 1)$, i.e., the random variable X has density

$$f_X(x) = \frac{1}{\pi(1+x^2)} \quad \text{for } x \in \mathbb{R}$$

- (a) Show that $\int_0^\infty xf(x) dx = \infty$ and $\int_{-\infty}^0 xf(x) dx = -\infty$.
 - (b) Why doesn't $\mathbb{E}[X]$ exist, even though $\int_{\mathbb{R}} xf(x) dx$ appears (naively) to be 0?
 - (c) Show that $\int_{\mathbb{R}} |x|f(x) dx = \infty$.
7. Consider the set $\Omega = \{0, 1\}^{\mathbb{N}}$, equipped with the σ -algebra $\mathcal{F}$ generated by cylinder sets (see below), and with the probability measure $\mu(\cdot) = \text{Leb}(D(\cdot))$.

Remark: For each finite sequence $(a_1, \dots, a_n) \in \{0, 1\}^n$, the cylinder set is

$$\xi_{a_1, \dots, a_n} := \{\omega \in \Omega : \omega_1 = a_1, \dots, \omega_n = a_n\}.$$

Then the σ -algebra $\mathcal{F}$ is the smallest σ -algebra containing all such cylinder sets, i.e.:

$$\mathcal{F} := \sigma(\{\xi_{a_1, \dots, a_n} : n \in \mathbb{N}, a_i \in \{0, 1\}\})$$

Equivalently, $\mathcal{F}$ is generated by the coordinate projection maps $\pi_n : \Omega \rightarrow \{0, 1\}$ where $\pi_n(\omega) = \omega_n$.

Let $E_n := \{\omega \in \Omega : \omega_n = 1\}$ and $A_n := \{\omega \in \Omega : \omega_n = 1, \omega_{n+1} = 0\}$. Clearly, $\{E_n\}_{n=1}^\infty$ are independent.

- (a) Show that $\{A_n\}_{n=1}^\infty$ are not independent.
 - (b) Prove that $\mathbb{P}[A_n \text{ i.o.}] = 1$.
 - (c) Let $A := \{\omega \in \Omega : \sum_{n=1}^\infty \omega_n < \infty\}$. Show that $A \in \mathcal{T}$, the tail σ -algebra associated with $\{E_n\}$. What is $\mathbb{P}[A]$?
 - (d) Let $s \in [0, 1]$. Define $A_s := \{\omega \in \Omega : \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{i=1}^n \omega_i = s\}$. Show that $A_s \in \mathcal{T}$ and that there is at most 1 value s_0 such that $\mathbb{P}[A_{s_0}] = 1$.
8. Let $\{X_n\}_{n \geq 0}$ be a sequence of random variables on a probability space $(\Omega, \mathcal{F}, \mathbb{P})$. Define the 'random power series' $s : \mathbb{R} \times \Omega \rightarrow \mathbb{R}$:

$$s(x, \omega) := \sum_{n=0}^{\infty} X_n(\omega) \cdot x^n,$$

and let $R(\omega)$ be the radius of convergence of $s(\cdot, \omega)$. Show that $R(\cdot)$ is constant a.s.

Hint: Recall that in $\mathbb{R}$, a power series $f(x) = \sum_{n=0}^\infty a_n x^n$ has radius of convergence given by

$$R = \frac{1}{\limsup_{n \rightarrow \infty} |a_n|^{\frac{1}{n}}}$$

9. Let $\{X_n\}_{n \geq 1}$ be a sequence of identically distributed random variables (not necessarily independent) on a probability space $(\Omega, \mathcal{F}, \mathbb{P})$ such that $\mathbb{E}[X_1^2] < \infty$. Show that for every $\varepsilon > 0$,

$$\lim_{n \rightarrow \infty} n\mathbb{P}[|X_1| \geq \varepsilon\sqrt{n}] = 0.$$

Hint: You will need Markov's inequality, but you will need something a little more.

10. Let $\{X_n\}_{n=1}^\infty$ be a sequence of random variables such that for every $\varepsilon > 0$,

$$\lim_{n \rightarrow \infty} \mathbb{P}\left\{\sup_{i \geq n} |S_i - S_n| > \varepsilon\right\} = 0$$

Show that there exists a random variable X such that $\lim_{n \rightarrow \infty} X_n = X$ a.s.

11. Give an example of a sequence of random variables $\{X_n, n \geq 1\}$ adapted on a filtered probability space $(\Omega, \mathcal{F}, \{\mathcal{F}_n\}, \mathbb{P})$ such that:

- (a) $\{X_n, n \geq 1\}$ is a martingale that converges a.s. but not in L^1
- (b) $\{X_n, n \geq 1\}$ satisfies $\sup_{n \geq 1} \mathbb{E}[|X_n|] < \infty$ but not $\limsup_{n \geq 1} X_n < \infty$ a.s.

12. Let $\{B_t, t \geq 0\}$ be a standard Brownian motion.

- (a) Define $I_t = \int_0^t B_s ds$. Find the mean and variance of I_t .
- (b) Show that $\mathbb{P}\{\sup_{t \in \mathbb{R}^+} B_t = \infty\} = \{\sup_{t \in \mathbb{R}^+} B_t > 0\} = \mathbb{P}\{\inf_{t \in \mathbb{R}^+} B_t = -\infty\} = 1$.
- (c) Show that

$$\mathbb{P}_0 \left\{ \bigcap_{n \in \mathbb{N}} \{\exists t \geq n : B_t = 0\} \right\} = 1$$

- (d) Show that if $\{Z_k\}_{k=0}^\infty$ are i.i.d. $N(0, 1)$ random variables and we define

$$X_t := \frac{t}{\sqrt{\pi}} Z_0 + \sqrt{\frac{2}{\pi}} \sum_{k=1}^\infty \frac{\sin kt}{k} Z_k$$

that then X_t is a Brownian motion.

Hint: Show that X_t and B_t have the same law, i.e., that $\mathcal{L}(X_t) = \mathcal{L}(B_t)$.

13. Let $\mathcal{Z}$ be the zero set of a standard one-dimensional Brownian motion $\{B_t, t \geq 0\}$, i.e., $\mathcal{Z} := \{t \geq 0 : B_t = 0\}$. Show that:

- (a) $\mathcal{Z}$ is a.s. a perfect set, i.e., it is a closed set with no isolated point.
- (b) $\text{length}(\mathcal{Z}) = 0$ a.s.
- (c) $\dim_H(\mathcal{Z}) \leq \frac{1}{2}$ a.s., where $\dim_H(E)$ is the Hausdorff dimension of $E \subseteq \mathbb{R}^d$.
(In fact, $\dim_H(\mathcal{Z}) = \frac{1}{2}$ a.s.)

Hint: prove first that $\exists c_0, c_1, s_0, 0 < c_0 < c_1 < \infty, s_0 > 0$ such that for all $t \in [1, 2]$ and $0 \leq s \leq s_0$:

$$c_0 \sqrt{s} \leq \arccos \sqrt{\frac{t}{t+s}} \leq c_1 \sqrt{s}$$

and that if $I = [t_0, t_1] \subseteq [1, 2]$ and $t_1 - t_0$ is small enough, then

$$\mathbb{P}_0\{\exists t \in I : B_t = 0\} \leq \frac{2}{\pi} c_1 \sqrt{\text{length}(I)}.$$

14. Let B^2 be a ‘complex Brownian motion’, so it is a two-dimensional Brownian motion using the complex representation of $\mathbb{R}^2$. In particular, $B^2 = B_1 + iB_2$ where (B_1, B_2) is a pair of independent standard one-dimensional Brownian motions. Let P be a non-constant polynomial with complex coefficients.

Prove that for any $\varepsilon > 0$, the compact set $\{z : |P(z)| \leq \varepsilon\}$ is non-empty (and therefore conclude that there is a solution to the equation $P(z) = 0$, i.e., the fundamental theorem of algebra).

Problems in statistics

15. Suppose that X_1 and X_2 are i.i.d. $\text{Exponential}(\theta)$ random variables.
- (a) If we observe only X_2 , show that the MLE of θ is $\hat{\theta} = X_2$.
 - (b) Suppose that we instead observe only $Z = (X_2 - 1)/X_1$. Find the joint distribution of (X_1, Z) , and integrate out X_1 to get the likelihood function.
 - (c) Suppose that $X_2 = 1$. Compare the MLEs for θ from parts (a) and (b).
16. Suppose that $\{X_i\}_{i=1}^n$ are i.i.d. $\text{Bernoulli}(p)$ random variables.
- (a) Show that the variance of the MLE of p attains the Cramér–Rao Lower Bound.
 - (b) For $n \geq 4$, show that the product $X_1 X_2 X_3 X_4$ is an unbiased estimator of p^4 .
 - (c) Find the best unbiased estimator of p^4 .
17. Observations $Y_1, \dots, Y_n$ are made according to the model $Y_i = \alpha + \beta x_i + \epsilon_i$, where $x_1, \dots, x_n$ are fixed constants and $\epsilon_1, \dots, \epsilon_n$ are i.i.d. $N(0, \sigma^2)$ random variables. Let $\hat{\alpha}$ and $\hat{\beta}$ denote MLEs of α and β .
- (a) Assume that $x_1, \dots, x_n$ are observed values of i.i.d. random variables $X_1, \dots, X_n$ with distribution $N(\mu_X, \sigma_X^2)$. Prove that when taking expectations over the joint distribution of X and Y , it is still the case that $\mathbb{E}[\hat{\alpha}] = \alpha$ and $\mathbb{E}[\hat{\beta}] = \beta$.
 - (b) Calculate the unconditional covariance of $\hat{\alpha}$ and $\hat{\beta}$. What can you say about it (c.f. part (a))?
Hint: use the joint distribution of X and Y .